

# End Semester Examination

MTH419

Number Theory

19 November 2025

## Question 1

Given a polynomial  $P(x) \in \mathbb{Z}[X]$  such that:

- $P(X) \bmod 5 = (X - 2)Q_1(X)$  in  $\mathbb{F}_5[X]$
- $P(X) \bmod 13 = (X - 3)Q_2(X)$  in  $\mathbb{F}_{13}[X]$

Find an integer  $n$  so that

$$P(X) \bmod 65 = (X - n)Q_3(X)$$

for a suitable  $Q_3(X)$  in  $\mathbb{Z}/\langle 65 \rangle$ .

**Solution:**

We note that  $2 \cdot 13 - 5 \cdot 5 = 1$ . It follows that if we take

$$n = 2 \cdot 2 \cdot 13 - 3 \cdot 5 \cdot 5 = -23$$

then  $n - 2$  is divisible by 5 and  $n - 3$  is divisible by 13.

It follows that  $P(n)$  is divisible by 5 and 13 by the given properties of  $P(X)$ ; hence it is divisible by 65.

Now  $P(X) - P(n)$  is divisible by  $X - n$ ; i.e.  $P(X) - P(n) = (X - n)Q(X)$  for some polynomial  $Q$  in  $\mathbb{Z}[X]$ . Putting  $Q_3(X) = Q(X) \bmod 65$ , we see that we have the required identity.

## Question 2

Give an example of an increasing sequence  $(a_n)$  of integers such that for every prime  $p$ , there is a subsequence which converges in  $|\cdot|_p$ .

**Solution:**

Consider the sequence defined by  $a_n = n$  which is clearly increasing.

For each prime  $p$ , the sequence  $(p^k)$  is a subsequence of this sequence that converges to 0 in  $p$ -adic topology.

Alternatively, for any integer  $m$  bigger than 1, consider the sequence defined by  $a_n = m^n$ . We have seen that for every prime  $p$ , the sequence  $m^{p^k}$  converges in  $p$ -adic topology.

### Question 3

For each  $n = 1, 2, 3, 4$  given an example of an irreducible polynomial of degree  $n$  in  $\mathbb{F}_3[X]$ .

**Solution:**

1. For  $n = 1$ , any polynomial of degree 1 is irreducible!
2. For  $n = 2$ , we note that  $-1$  is not a square in  $\mathbb{F}_3$ . Thus  $X^2 + 1$  is irreducible in  $\mathbb{F}_3[X]$ .
3. For  $n = 3$ , we note that every element of  $\mathbb{F}_3$  is a root of  $X^3 - X$ . It follows that  $X^3 - X - 1$  has no root in  $\mathbb{F}_3$ . Since this is a cubic polynomial, it is irreducible.
4. For  $n = 4$ , we note that  $1, 3, 3^2 = 9, 3^3 = 27, 3^4 = 81$ . So 3 has order 4 modulo 5. This means that if  $\zeta$  is a primitive 5-th root of unity, the  $\zeta$  lies in  $\mathbb{F}_{3^4}$  and no smaller field.

It follows that the polynomial  $X^4 + X^3 + X^2 + X + 1 = (X - \zeta)(X - \zeta^2)(X - \zeta^3)(X - \zeta^4)$  is irreducible in  $\mathbb{F}_3$ .

### Question 4

For which of the following values of  $P(X)$  is the ring  $\mathbb{Q}[X]/\langle P(X) \rangle$  a field?

1.  $P(X) = X^3 - 5X + 1$
2.  $P(X) = X^3 - 2X + 1$
3.  $P(X) = X^3 - 4X + 6$
4.  $P(X) = X^4 - X^3 + X^2 - X + 1$

**Solution:**

Asking if this ring is a field is the same as asking whether the polynomial is irreducible in  $\mathbb{Q}[X]$ . Since each of these is a monic polynomial with integer coefficients, this is true if and only if it is irreducible in  $\mathbb{Z}[X]$ .

1. We note that if  $X^3 - 5X + 1$  is reducible in  $\mathbb{Z}[X]$ , then 1 or  $-1$  must be a root. Since neither of them is a root, we see that  $X^3 - 5X + 1$  is irreducible.
2. Substituting  $X = 1$  we get 0. Hence  $X^3 - 2X + 1 = (X - 1)Q(X)$  for some quadratic polynomial  $Q$ . So the given polynomial is reducible.
3. The given polynomial satisfies Eisenstein's criterion for irreducibility with respect to the prime 2. Thus, it is irreducible.

4. In the earlier question we have seen that  $X^4 + X^3 + X^2 + X + 1$  is irreducible modulo 3; hence it is irreducible in  $\mathbb{Z}[X]$ . Substituting  $X$  by  $-X$ , we see that  $X^4 - X^3 + X^2 - X + 1$  is irreducible also.

Alternatively, we substitute  $X = Y - 1$  to get  $Y^4 - 5Y^3 + 10Y^2 - 10Y + 5$  which is an Eisenstein polynomial with respect to the prime 5. Hence, the polynomial is irreducible.

### Question 5

Consider the subfield  $K = \mathbb{Q}[\sqrt{2}, \sqrt{-1}]$  of  $\mathbb{C}$ .

1. Give a basis of  $K$  as a vector space over  $\mathbb{Q}$ .
2. What is  $\text{Tr}_{K/\mathbb{Q}}(\sqrt{2})$ ?
3. What is  $\text{Nm}_{K/\mathbb{Q}}(\sqrt{-2})$ ?
4. Given an example of an element  $\alpha$  of  $K$  whose minimal polynomial is of degree 4.

**Solution:**

We note that  $K$  is a quotient of  $\mathbb{Q}[X, Y]/\langle X^2 - 2, Y^2 + 1 \rangle$ . So every element of  $K$  can be written as a polynomial in  $X$  and  $Y$  with degrees at most 1 in each variable. Thus,  $1, X, Y, XY$  is a spanning set for  $K$  over  $\mathbb{Q}$ . This corresponds to the set  $S = \{1, \sqrt{2}, \sqrt{-1}, \sqrt{-2}\}$  of elements of  $\mathbb{C}$ .

We note that

$$M_S(\sqrt{2}) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 2 & 0 \end{pmatrix}$$

while

$$M_S(\sqrt{-1}) = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}$$

Thus, we have

$$M_S(\sqrt{-2}) = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 2 & 0 \\ 0 & -1 & 0 & 0 \\ -2 & 0 & 0 & 0 \end{pmatrix}$$

By looking at the first row (or column), we see that  $M_S(1), M_S(\sqrt{2}), M_S(\sqrt{-1}), M_S(\sqrt{-2})$  are linearly independent. Thus,  $S$  is linearly independent over  $\mathbb{Q}$ .

Using these matrices we observe that:

- $\text{Tr}_{K/\mathbb{Q}}(\sqrt{2}) = 0$
- $\text{Nm}_{K/\mathbb{Q}}(\sqrt{-2}) = 4$

Consider the element  $\alpha = (\sqrt{2} + \sqrt{-2})/2$ . We calculate that  $\alpha^2 = \sqrt{-1}$ . Thus,  $\alpha$  satisfies  $X^4 + 1$ .

Since this is an irreducible polynomial over  $\mathbb{Q}[X]$ , we see that it is the minimal polynomial of  $\alpha$ .

### Question 6

Given the polynomial  $P(X) = X^3 + 2X + 6$  in  $\mathbb{Z}_3[X]$ .

What is the structure of  $\mathbb{Z}_3[X]/\langle P(X) \rangle$ .

**Solution:**

We first consider the polynomial modulo 3 to get  $X^3 + 2X$ . This has three distinct roots in  $\mathbb{F}_3$ . In fact,  $X^3 + 2X = X(X + 1)(X + 2)$  in  $\mathbb{F}_3[X]$ . These are distinct irreducible factors.

Thus, by results proved in class, we see that

$$\frac{\mathbb{Z}_3[X]}{\langle P(X) \rangle} \cong \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_3$$

More explicitly, we can use Hensel's lemma to lift the roots  $0, 1, 2$  in  $\mathbb{F}_3$  to roots  $\eta_0, \eta_1$  and  $\eta_2$  in  $\mathbb{Z}_3$  so that  $P(X) = (X - \eta_0)(X - \eta_1)(X - \eta_2)$  in  $\mathbb{Z}_3$ . The decomposition above follows.

### Question 7

Observe that  $P(X) = X^3 - 2X + 6$  has the root 1 in  $\mathbb{F}_5$ .

1. Find an element  $n$  of  $\mathbb{Z}/\langle 25 \rangle$  which is a root of  $P(X)$  in that ring.
2. Show that  $P(X) = (X - \alpha)Q(X)$  in the  $\mathbb{Z}_5[X]$  where  $Q(X)$  is irreducible in  $\mathbb{Z}_5[X]$ .

**Solution:**

We note that  $P'(X) = 3X^2 - 2$  takes the value  $P'(1) = 1$ . By Hensel lifting, we have

$$x_{n+1} = x_n - P(x_n)$$

starting with  $x_0 = 1$ . Thus,  $x_1 = 1 - P(1) = 1 - 5 = -4$ . We let  $\alpha$  be the limit of  $(x_n)$  in  $\mathbb{Z}_5$ .

We now calculate  $Q(X) = P(X)/(X - \alpha)$  in  $\mathbb{Z}_5$  by calculating it modulo  $5^2$ .

We divide  $P(X)$  by  $X + 4$  to get

$$P(X) = (X + 4)(X^2 - 4X + 14) - 50$$

so that  $Q(X) \bmod 5^2 = X^2 - 4X + 14$ .

We further note that

$$X^2 - 4X + 14 = (X - 2)^2 + 10$$

and  $Y^2 + 10$  is an Eisenstein polynomial in  $\mathbb{Z}/\langle 5^2 \rangle$ . It follows that  $Q(X) \bmod 5^2 = X^2 - 4X + 14$  is irreducible over  $\mathbb{Z}/\langle 5^2 \rangle$ . Thus,  $Q(X)$  is itself irreducible.

### Question 8

Consider the ring  $S = \mathbb{Z}[X]/\langle X^3 - 4X + 1 \rangle$  as a lattice in  $A = \mathbb{R} \otimes_{\mathbb{Z}} S$ .

1. Show that  $A$  is isomorphic to  $\mathbb{R} \times \mathbb{R} \times \mathbb{R}$  as an  $\mathbb{R}$ -algebra.
2. Using Minkowski's theorem what is the smallest  $r > 0$  such that the cube  $\{(x_1, x_2, x_3) : |x_i| \leq r\}$  contains a non-zero element of  $S$ .

**Solution:**

We calculate the discriminant of  $P(X) = X^3 - 4X + 1$  as  $-4(-4)^3 - 27(1)^2 = 256 - 27 = 229$ .

Since the discriminant is a square in  $\mathbb{R}$  for a cubic polynomial, it follows that all three roots are in  $\mathbb{R}$ .

A  $\mathbb{Z}$ -basis of  $S$  is given by the image of  $\{1, X, X^2\}$  in  $S$ . Thus, let  $w_1, w_2, w_3$  be the real roots of the above equation. Then the image of  $S$  in  $\mathbb{R}^3$  is given by

$$\begin{pmatrix} 1 & w_1 & w_1^2 \\ 1 & w_2 & w_2^2 \\ 1 & w_3 & w_3^2 \end{pmatrix} : \mathbb{Z}^3 \rightarrow \mathbb{R}^3$$

The volume of the quotient is the absolute value of the determinant of this matrix which is  $|(w_1 - w_2)(w_2 - w_3)(w_3 - w_1)|$  which is the square root of the discriminant 229.

The volume of the *compact* cube given above is  $8r^3$ .

In order to have a lattice point, Minkowski's theorem requires the volume of a compact, convex, symmetric region to be at least  $2^n$  times the volume of quotient of  $\mathbb{R}^n$  by the lattice.

Thus, we need  $8r^3 \geq 2^3 \sqrt{229}$  which gives  $r \geq (229)^{1/6}$ .

**Note:** It can be observed that we have the point  $(1, 1, 1)$  in the lattice. So Minkowski's bound is not optimal. To get a more useful bound out of Minkowski's theorem, we can apply it to the lattice of trace 0 elements of  $S$  inside the trace 0 elements of  $A$ .